

HARDWARE DOCUMENTATION FOR OSHWA CERTIFICATION

HOS Trigonometry 1.5

Humanoid Operating System · robotic-arm Cartesian layer only · physical joystick command

Arduino Uno · CNC Shield V3 · A4988 drivers · analog joystick · NEMA 17 steppers · time-of-flight · IR grip sensor

Name: **HOS Trigonometry 1.5**. Version: **1.5**. Scope: the robotic-arm Cartesian layer only. It is not the mobile base, not the vision stack, and not Shell 1.5. It belongs to the Humanoid Operating System published at github.com/Aarav-111/Humanoid-Operating-System by Prolabs Robotics (project lead Aarav Jaisingh, co-lead Siyona Chicker). HOS splits a robotic arm (the worker) from a mobile base (the mover). Trigonometry 1.5 is only the arm: a three-axis Cartesian mechanism that carries the tool through X, Y and Z.

This file is the hardware document for that version. It is written so another builder can study, modify, and reproduce the arm. It is not itself an OSHWA certificate. Certification is a separate submission at certification.oshwa.org once this document and the design files are public.

What is original, and what is not. Arduino Uno, CNC Shield V3, A4988 modules, the analog joystick breakout, NEMA 17 motors, the metal 10 A supply, belts, pulleys, and the leadscrew are third-party modules with public datasheets. The creator contribution is the Cartesian integration, the lid-mounted physical command console, the hybrid belt/leadscrew drive, and Trigonometry 1.5, the firmware that resolves a joystick vector or an HOS grid cell into step pulses.

1. What HOS Trigonometry 1.5 is

HOS separates a robotic arm from a mobile base. This document covers the arm only. Trigonometry 1.5 is that arm in Cartesian form: belts on X and Y, a leadscrew on Z, a tool motor on the carriage, and a joystick that jogs those three axes. Locomotion, navigation, and the camera planner are other HOS layers and are out of scope here.

X and Y are belt-driven on aluminium V-slot. Z is a leadscrew with parallel smooth-rod guidance. Motion electronics are an Arduino Uno, a CNC Shield V3, and A4988 drivers, fed by a separate 10 A metal supply. An HOS plan can ask this arm for a grid move — `goto_coordinate(T, 8)`, `pickup()`, `keep()` — but the trigonometry that executes it lives on this stage: column and row become millimetres, a joystick deflection becomes a vector, and the vector becomes STEP/DIR on the arm axes only. Two sensors close that loop on the arm itself. A time-of-flight sensor on the carriage reads distance to the ground, so Z is a measured height, not only a step count. IR sensors are mounted horizontally across the jaw gap, so the beam sees an object before the sponge closes on it.

The operator does not have to sit at a laptop to move the arm. The analog joystick on the controller lid is the hand control for teach and jog of X, Y and Z.

2. What is different

A stock GRBL plotter or mini-mill is a PC peripheral. This build adds a command layer that lives on the machine.

Physical software. The joystick is not a decoration. Its two potentiometers and switch are read by the Arduino. Deflection becomes a signed step rate on X and Y; the switch selects Z, enables the drivers, or latches a mode. The same STEP/DIR pins the shield already wires to the A4988s carry those pulses. One firmware owns both the human command and the motor timing. That is the “physical software command” layer: the control law is software, the affordance is a hand control.

Hybrid axes, on purpose. Belts on X and Y give speed and low moving mass for jogging across the table. The Z leadscrew gives holding torque and finer pitch when the tool is loaded vertically. A pure-belt Z would back-drive. A pure-screw XY would be slow to teach. The split matches the job.

Console on the power brick. The controller is not a loose breadboard. The Uno and shield sit in a white printed (or foam-board) bay, the joystick is fixed to the lid, and that bay sits on the metal supply. Power, logic, and the hand control are one station at the corner of the frame.

A tool motor on the carriage. A horizontal NEMA 17 on the Z carriage, separate from the vertical leadscrew motor, carries the end effector. That actuator is part of the arm, not a fourth positioning axis. The shield's A socket drives it.

Ground distance and grip presence. A time-of-flight sensor is aimed down from the carriage and reports millimetres to the ground or table. That reading is the arm's Z truth: jog and grid moves stop at a height floor instead of trusting the leadscrew count alone. An IR sensor is mounted horizontally at the gripper, beam across the jaw gap, and returns present or absent. The jaw face is a sponge, for a compliant grip. `pickup()` is allowed only when the horizontal IR sees an object. An empty close is refused.

3. What is well made

The frame is commodity 2020-class V-slot, joined by white printed brackets rather than machined plates. That is the right call for a reproducible student or farm-lab machine: a builder with a printer and a hacksaw can copy the nodes. Motor mounts, belt anchors, and the Z upright are the same family of parts, so a broken bracket is a reprint, not a shop job.

Motor power and logic power are split. The metal supply (label: 10 A, Keenonic-class enclosed SMPS) feeds the shield's VMOT. The Uno is powered over USB. A4988 modules are socketed, so a cooked driver is a swap. Z is constrained by the screw and two smooth rods, which keeps the carriage from twisting when the horizontal tool motor and its cable weight hang off the front.

Cable looms are zip-tied at the carriage and taped in bundles at the console. It is a working prototype, not a harnessed product: Dupont jumpers are still visible at the end effector, and the console bay is open at the front for probing. That openness is useful while tuning current limits. For a released version, the same pinout should move to a short shielded cable and a closed lid with a strain-relieved exit.

4. The innovation, stated plainly

The innovation is not a new motor or a new board. It is the closed loop between a hand and a three-axis Cartesian arm, with two local sensors in that loop. Teach and jog are physical. Height is measured by time of flight to the ground. A grip is a sponge on the shaft, and it is allowed only when the horizontally mounted IR sensor sees an object. The command path stays on the arm: potentiometer, ADC, dead-zone, step timer, A4988, NEMA 17, plus an I2C range reading and a digital object bit.

5. AprilTag navigation, coordinate design, and custom offset

Navigation on this arm is an AprilTag 36h11, read by S1.5-SRC. The tag is the fiducial. Its detected centre is not yet the board coordinate: the tag sits above the board, so perspective pulls it away from the camera nadir. Trigonometry 1.5 is that correction, plus the fixed cell offset from the tag to the gripper.

The working grid is 20 by 20 by default, square cells, columns A–T and rows 1–20, on a board about 24 inches wide. A cell name such as G11 is column G, row 11. Row 1 is the top of the image. Column A is the left. The gantry runs one axis at a time: all vertical steps first, then all horizontal steps. The serial letters are u, d, l, r, and s for stop.

Heights are measured above the board. The shipped Trigonometry panel values are camera height $H = 35.0$ in, tag height $h = 5.5$ in, board height 0. A point at height h and horizontal offset r from the nadir images at the same angle as a flush point at $r \times H / (H - h)$. The true board position is the raw pixel pulled back toward the nadir by $k = (H - h) / H$. With these heights, $k = 29.5 / 35 = 0.843$. If the camera is not above the board, or the tag is at or above the camera, k falls back to 1 and no correction is applied.

The nadir is the centre of the grid box plus a pixel pivot. Pivot X and pivot Y ship at 0. For a raw tag centre (cx, cy) and nadir (ccx, ccy) : corrected = $(ccx + k(cx - ccx), ccy + k(cy - ccy))$. That corrected point is the tag's cell. There is no second fitted homography; the panel formula is the correction the board uses.

The gripper is not on the tag. The custom offset is a cell delta from the tag cell to the gripper cell, capped at 10 cells on each axis. The shipped offset is 7 cells up and 0 cells right/left (up/down = -7 , right/left = 0; up is a negative row step because row 1 is the top). A plan coordinate is the working point. The arm is sent to the tag cell that puts the gripper on that point.

A pre-action nudge trims the last approach. It ships at 0.2 s to the right, armed before pickup and before release, off before keep and press. A contact token from the arm — a whole serial token s, S, or stop — ends a descent. It is not matched as a letter inside another word.

The time-of-flight reading remains the ground distance for the height floor. The IR bit remains the object-present test at the jaws. The AprilTag does not replace either sensor. It names the cell. The offset names where the gripper is relative to that cell.

6. Hardware views

The views below are the HOS Trigonometry 1.5 hardware: the Cartesian stage, the joystick console on the 10 A supply, the Z carriage with the tool motor, and the printed-part family.

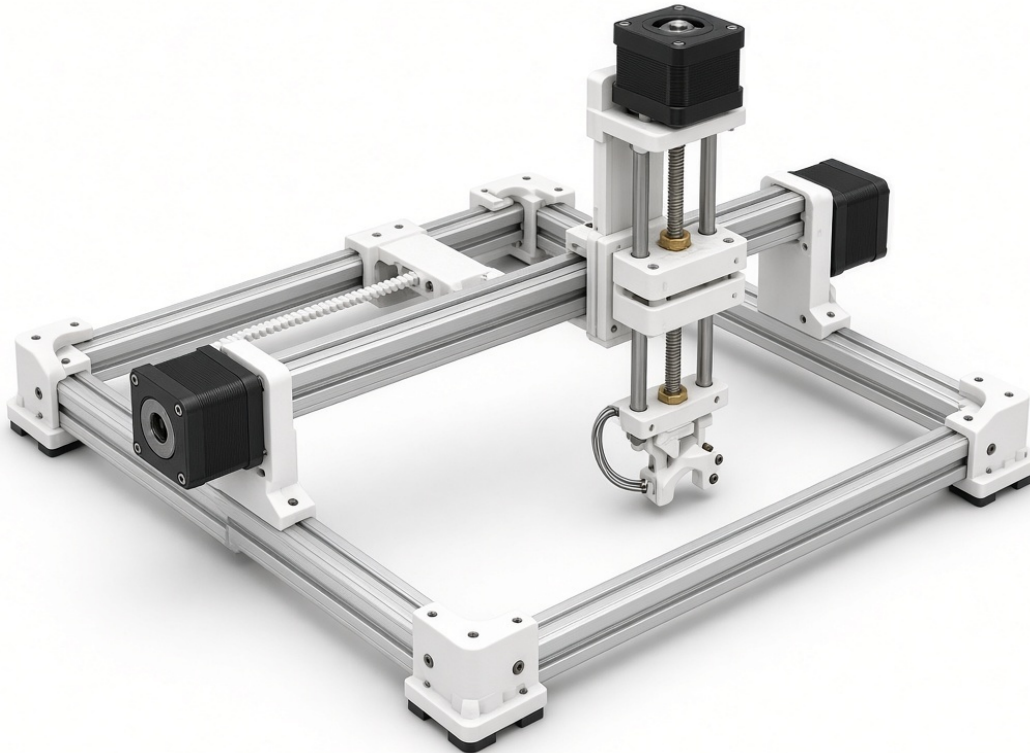


Figure 1. HOS Trigonometry 1.5 stage: belt X/Y, leadscrew Z, NEMA 17s, printed nodes.

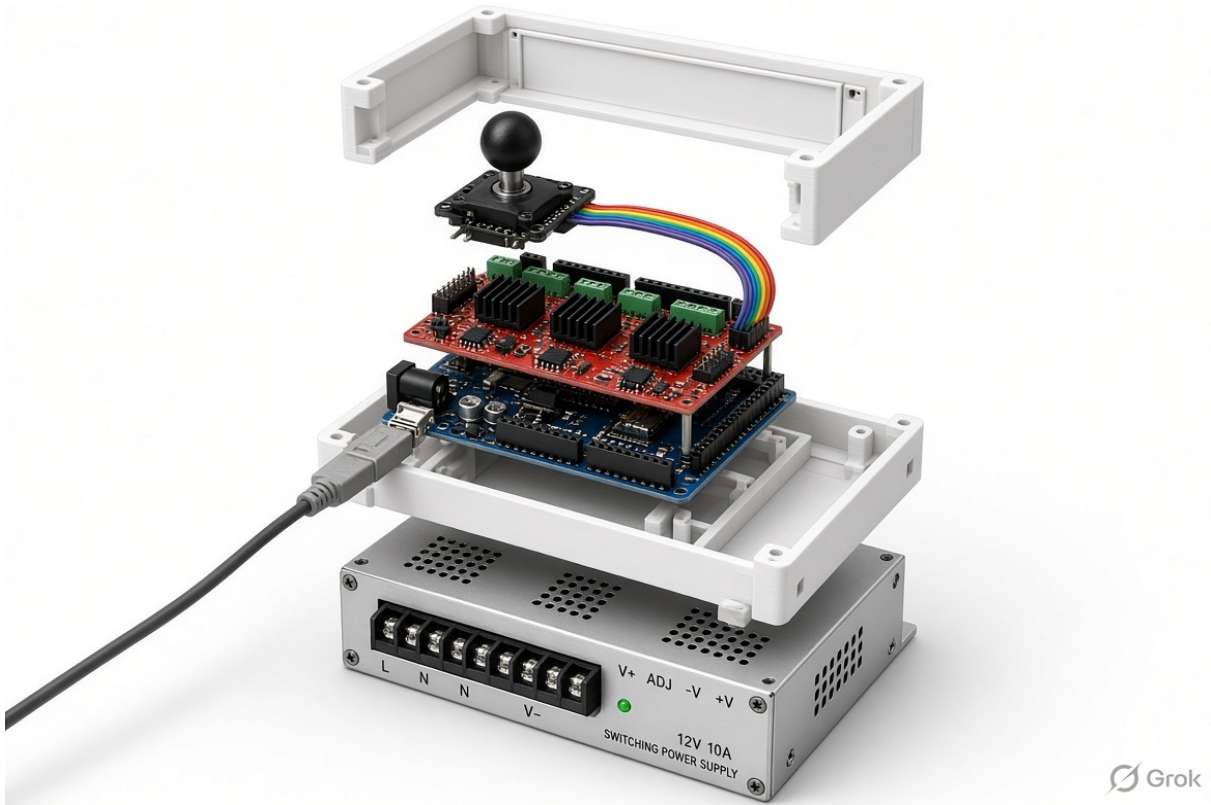


Figure 2. Command console: joystick lid, CNC Shield V3, Arduino Uno, 12 V 10 A supply.

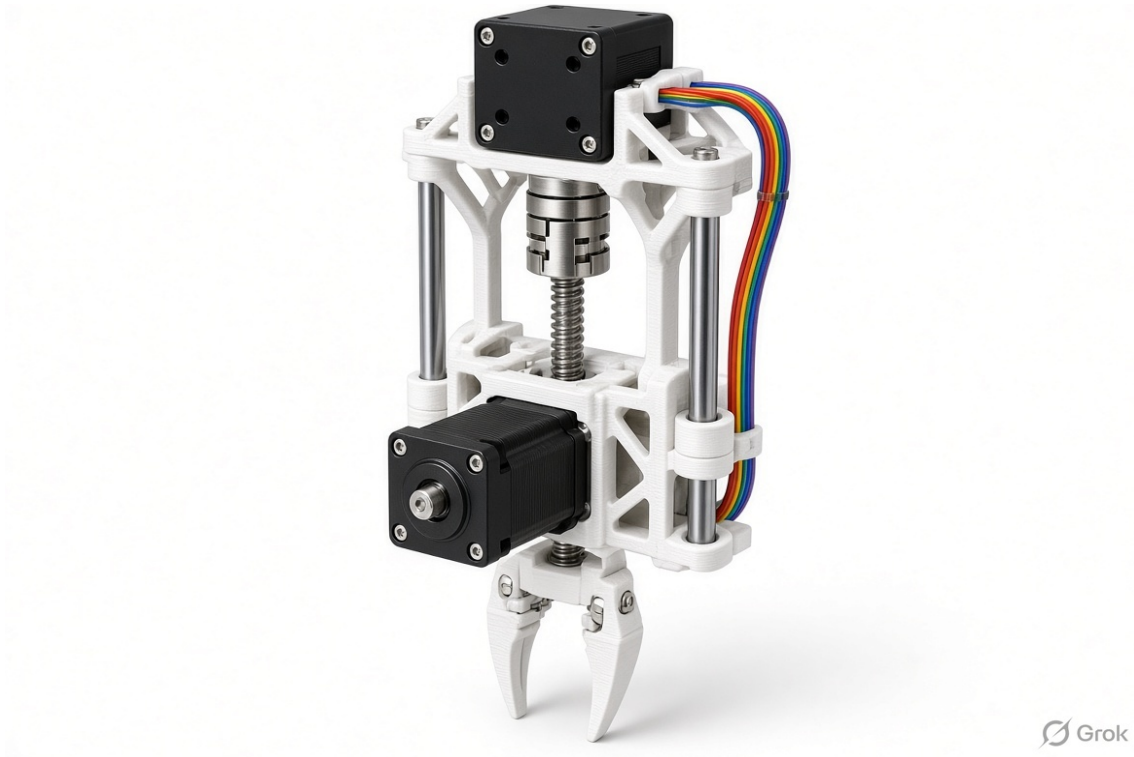


Figure 3. Z carriage: leadscrew motor, coupling, twin rods, tool motor, end effector.

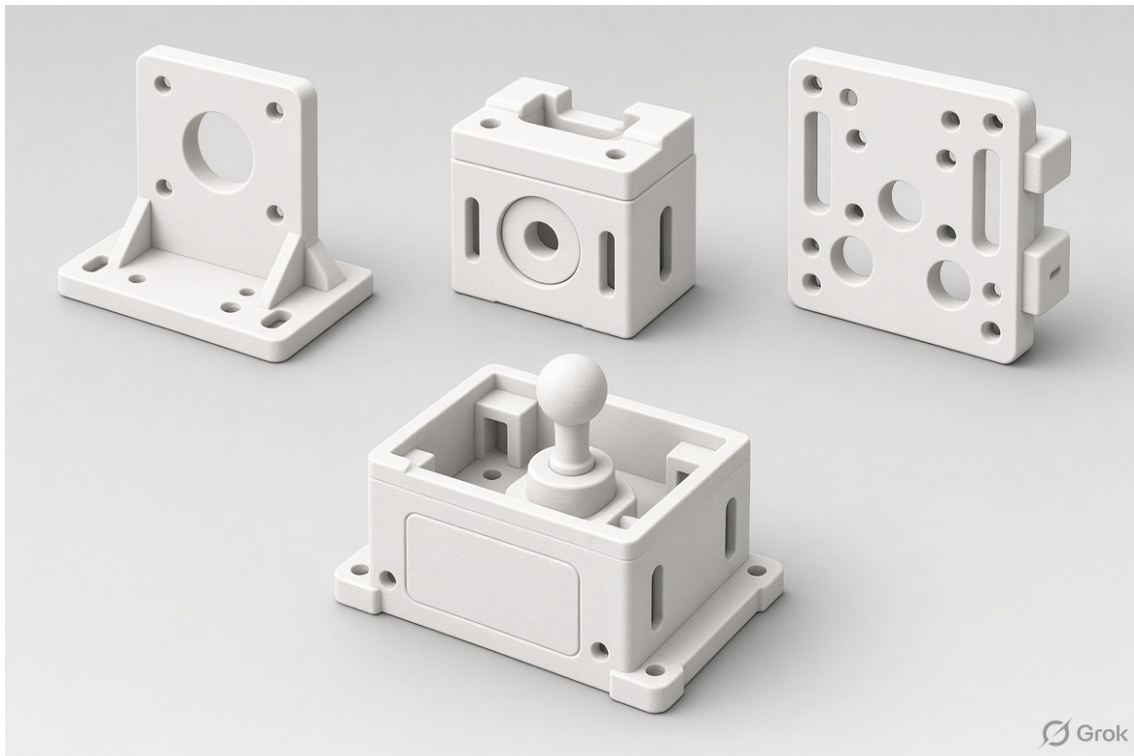


Figure 4. Printed parts: motor foot, idler block, carriage plate, joystick bay.

7. Machine record

The record below is the Trigonometry 1.5 arm: extrusion table, belt span, Z screw and smooth rod, and the joystick console at the corner of the frame.



Figure 5. HOS Trigonometry 1.5 arm. Extrusion table, belt span, Z screw and rod, joystick console.



Figure 6. Z upright, leadscrew, carriage with horizontal NEMA 17, end-effector loom.

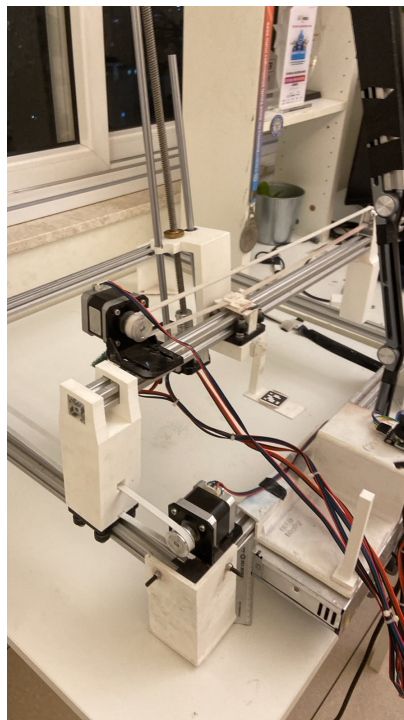


Figure 7. Belt axes. Two NEMA 17s with pulleys, printed feet and idlers, GT2 belt.

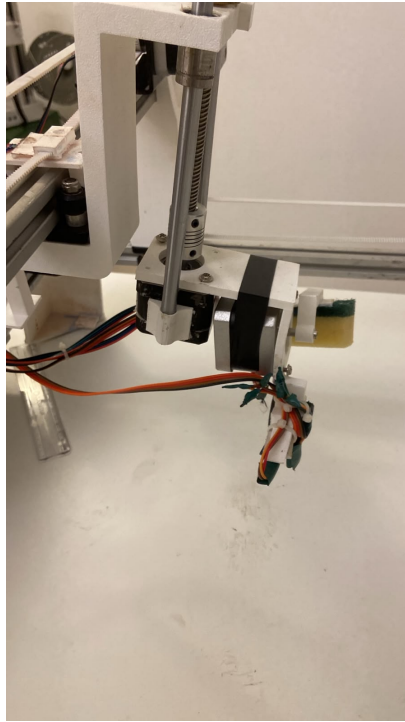


Figure 8. Carriage. Screw coupling, printed clamp, horizontal tool motor, tool loom.

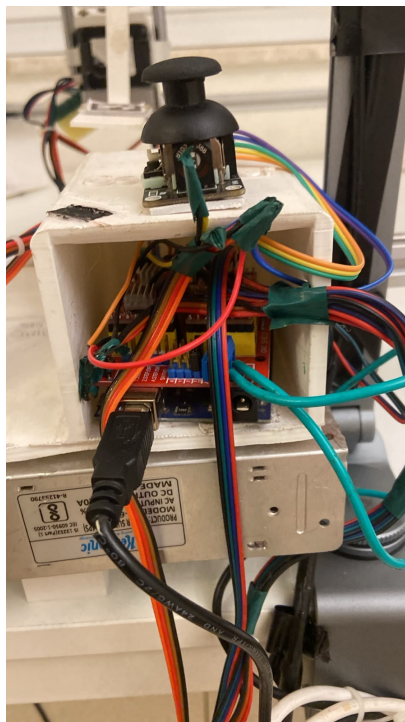


Figure 9. Command console. Joystick on the lid, Uno under the CNC shield, USB, 10 A supply.

8. Printed-part drawings

These are the 3D drawings of the printed parts. The sponge is the grip face: gripper part 2 couples it to the shaft. The IR sensors mount horizontally on gripper part 1, beam across the sponge gap, so an object in the jaws breaks or returns the beam before the sponge closes. The time-of-flight sensor mounts on the same part and faces the ground. The AprilTag connector holds the 36h11 tag used for navigation. The hub holds the Arduino, CNC Shield V3, A4988 drivers, and the supply.

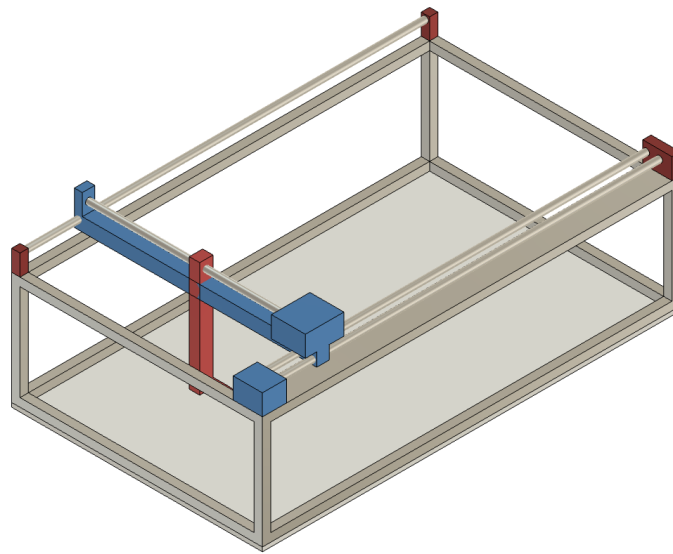


Figure 10. Basic robot drawing. Full Cartesian arm.

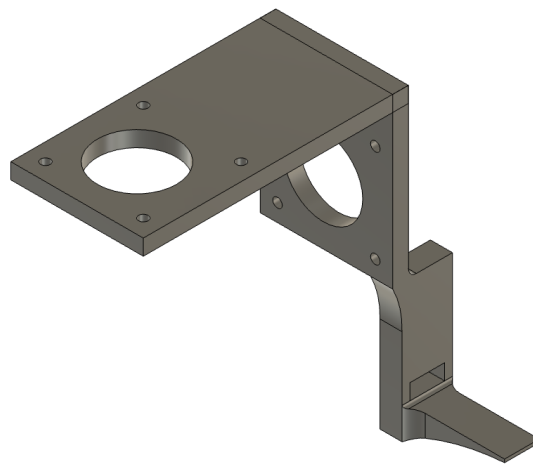


Figure 11. Gripper part 1. Joins the Z axis and the gripper stepper, and mounts the horizontal IR sensors and the time-of-flight sensor.



Figure 12. Gripper part 2. Couples the sponge to the shaft.

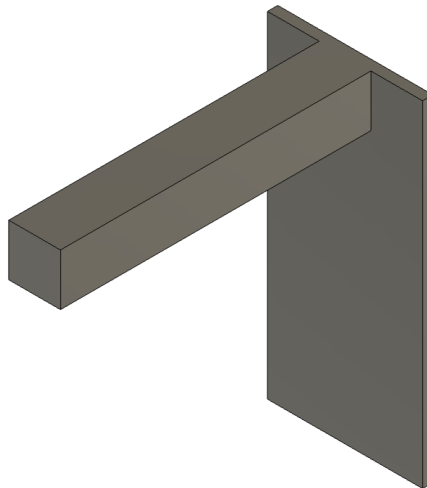


Figure 13. AprilTag connector. Holds the tag used for navigation.

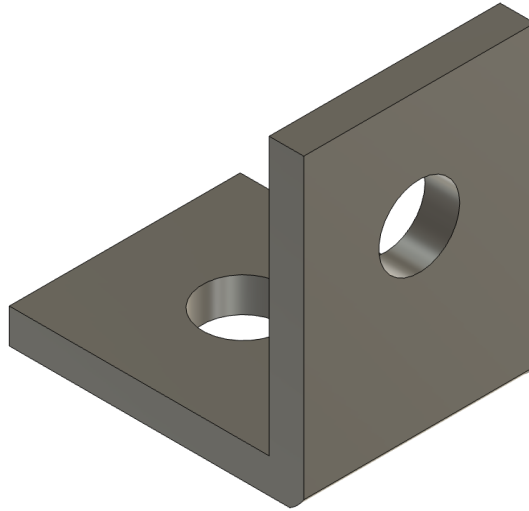


Figure 14. L-bracket. Connects the gripper to the robot.

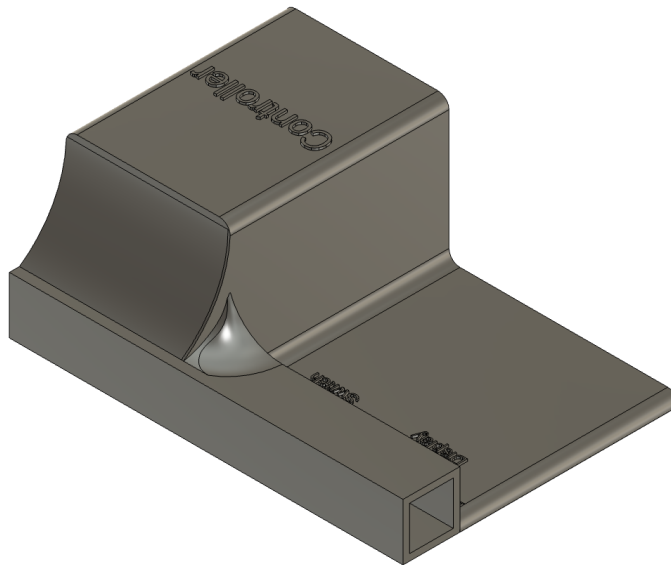


Figure 15. Hub. Holds the controller, supply, Arduino Uno, CNC Shield V3, and A4988 drivers.

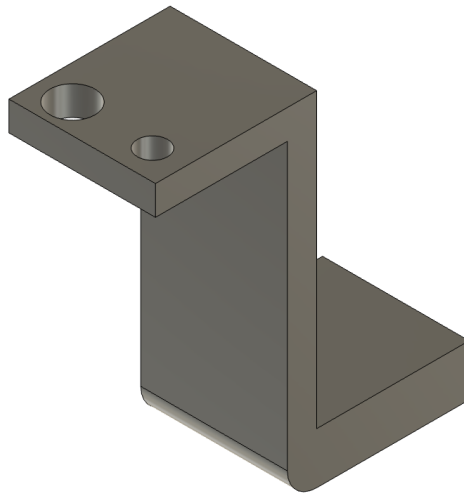


Figure 16. Z-axis mounter. Mounts the Z stepper on the Y axis.

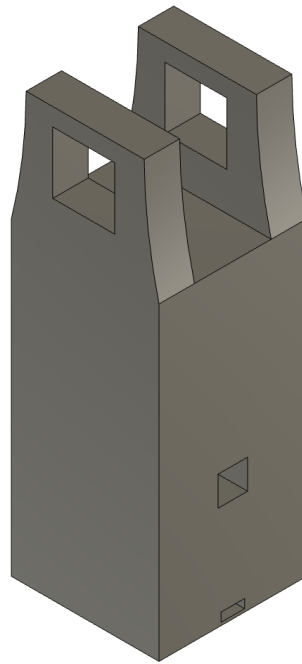


Figure 17. XY inter-axis connector. Joins the Y axis to the X-axis belts and V-slot carriage.

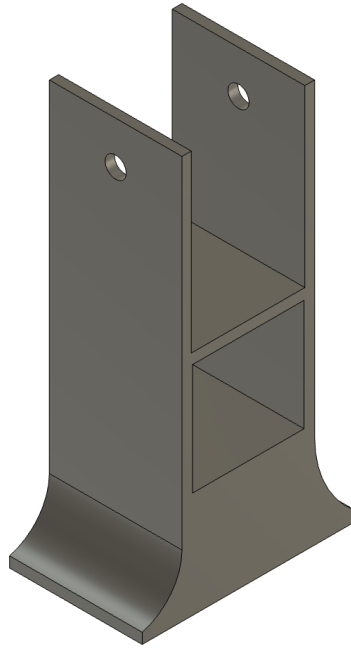


Figure 18. Y-axis holder. Supports the far side of the Y axis.

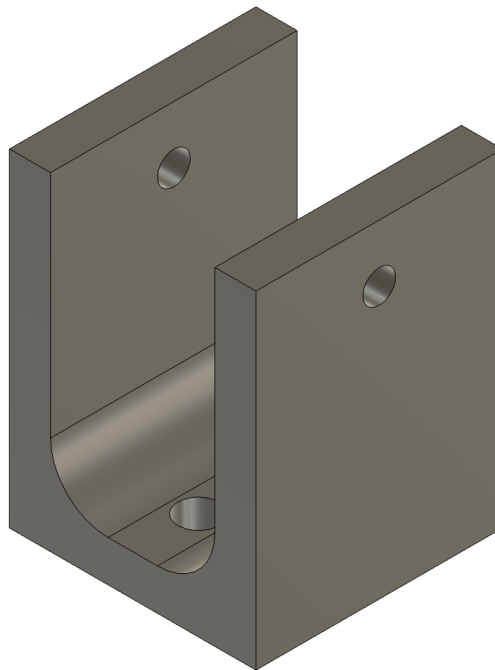


Figure 19. Pulley mount. Holds the pulleys on the axis.

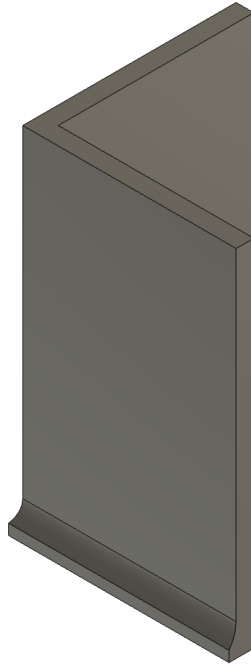


Figure 20. Legs. Hold the robot up from the ground.

9. Bill of materials

Quantities follow the named stack. Confirm before ordering. Third-party lines are not relicensed by this document. The 3MF files for the printed parts are released with this document.

Item	Qty	Role	Status
Arduino Uno (ATmega328P)	1	Command MCU, ADC, step timing	Third-party module, public datasheet
CNC Shield V3	1	STEP/DIR/EN breakout, VMOT, driver sockets	Third-party; open Gerbers exist from several makers
A4988 stepper driver	3 or 4	X, Y, Z, and optional A / tool	Third-party module, Allegro datasheet
Analog joystick module	1	Physical X/Y command and mode switch	Third-party breakout
NEMA 17 stepper, 4-wire	3 or 4	X, Y, Z, optional tool axis	Third-party actuator
Enclosed SMPS, 12 V class, 10 A	1	VMOT only. Label reads 10 A, Keenonic	Third-party. Do not copy the supply internals
2020 V-slot extrusion	frame	Table and gantry rails	Commodity. Cut list is creator design
GT2 belt, 6 mm, and pulleys	2 axes	X and Y transmission	Commodity
T8 leadscrew, nut, coupling	1	Z transmission	Commodity
Smooth rod, 8 mm class, and bearings	2	Z anti-rotation	Commodity
Printed brackets, bay, carriage	set	Creator structure	Original. Release STL/STEP
Time-of-flight sensor	1	Distance from the carriage to the ground	Third-party I2C breakout, VL53L0X class
IR object sensor	1	Mounted horizontally across the sponge gap	Third-party digital IR module
Sponge grip face	1	Compliant jaw on the shaft, via gripper part 2	Consumable
AprilTag 36h11	1	Fiducial for cell navigation and gripper offset	Printed tag, read by S1.5-SRC
USB cable, Dupont, ferrules	lot	Logic, sensor, and motor wiring	Commodity

A 10 A, 12 V supply is enough for three or four NEMA 17s if each A4988 Vref is set for about 1 A or less. Do not run the motors from the Arduino 5 V pin.

10. Electronics and pin map

The shield plugs onto the Uno. A4988 boards sit in the X, Y, Z sockets, and in A if the carriage tool motor is used. Orient each driver so the enable pin matches the shield silkscreen. Set the current-limit potentiometer before the first move: $V_{ref} \approx I_{max} \times 8 \times R_{sense}$, with R_{sense} typically 0.1 ohm on clone boards, so 1 A wants about 0.8 V. Confirm the sense resistor on the module in hand.

Signal	Uno / shield	Connects to
X STEP / DIR	D2 / D5	A4988 X, then X NEMA 17
Y STEP / DIR	D3 / D6	A4988 Y, then Y NEMA 17
Z STEP / DIR	D4 / D7	A4988 Z, then Z NEMA 17
A STEP / DIR	D12 / D13	Optional tool motor on carriage
Enable (active low)	D8	All A4988 EN, tied on the shield
Joystick VRx	A0	X deflection
Joystick VRy	A1	Y deflection, or Z when the switch is held
Joystick SW	A2	Mode: jog XY / jog Z. INPUT_PULLUP
ToF SDA / SCL	A4 / A5	I2C distance to ground. Keep these off the joystick
IR OUT	D11	Object present at the gripper, active low or high per module
VMOT / GND	Shield power term.	12 V supply. Logic GND common with Uno
5 V / GND	Shield 5 V header	Joystick and IR. ToF breakout from 5 V or 3.3 V as the module requires

Stock GRBL claims D2–D13 and A0–A2. This arm sketch is separate, not a GRBL patch. A4 and A5 stay on I2C for the time-of-flight sensor. Do not put the joystick there. Do not drive STEP from two firmwares at once.

11. Trigonometry 1.5 command layer

This layer accepts two arm commands and nothing else. A grid step `goto_coordinate(column, row)` maps onto this Cartesian arm: X = column pitch, Y = row pitch, Z = tool height. A hand on the stick is the teach input for the same three axes. The two potentiometers are a vector in the arm plane. Magnitude is the hypotenuse, $r = \sqrt{v_x^2 + v_y^2}$, and that magnitude sets the step interval. Direction is the sign of each component. Holding the switch replaces Y with Z, so the arm rises or lowers on the leadscrew. Z motion also watches the time-of-flight reading and will not drive through the height floor. The tool motor closes only if the IR sensor reports an object. The resolver does not drive a mobile base. That resolver is Trigonometry 1.5.

Behaviour as designed for this machine:

- Stick X deflects the X belt. Stick Y deflects the Y belt. Rate grows with deflection, direction follows the sign.
- Switch held: stick Y drives Z on the leadscrew instead of Y. X can stay live or lock, builder's choice; the as-documented default locks X while Z is selected, so a vertical move does not walk the gantry.
- A downward jog stops if the time-of-flight distance to ground is at or under the floor, about 20 mm until calibrated.
- `pickup()` runs the tool motor only when the horizontal IR sensor reads an object in the sponge gap. No object, no close.
- No host is required to jog. USB is for power and for uploading the sketch.

Reference jog loop, released with this document:

```
center = 512; dead = 40; floorMm = 20
vx = analogRead(A0) - center
vy = analogRead(A1) - center
if (abs(vx) < dead) vx = 0
if (abs(vy) < dead) vy = 0
zMode = (digitalRead(A2) == LOW)
groundMm = tof.readRangeMillimeters()
objectThere = (digitalRead(11) == LOW)
if (zMode and vy < 0 and groundMm <= floorMm) vy = 0
if (pickup requested and not objectThere) skip the tool motor
intervalX = map(abs(vx), dead, 512, 4000, 250) // microseconds
on each elapsed interval: DIR pin, then STEP high 5 us, then low
```

Tune the 250 μ s floor against the motor. A NEMA 17 on a GT2 belt will skip if the interval collapses too far, especially with the carriage cable drag. Start slower, then raise rate.

12. Mechanical notes from the build

X and Y. Each axis is a NEMA 17, a toothed pulley, a belt, a printed motor foot, and a printed idler or belt clamp. The belt runs along the extrusion. Direction is whatever the pulley winding makes it; swap one coil pair or the DIR invert in firmware if the stick and the table disagree.

Z. A vertical trapezoidal screw, coupling, and brass or POM nut carry the carriage. Two smooth rods stop the nut from spinning. The Z motor is mounted on the moving carriage rather than at the top of a fixed screw — a valid arrangement if the screw is fixed to the upright and the motor-nut unit climbs it. Match the coupling style in Figure 8 before copying the printed clamp.

Tool and sensors. The horizontal motor drives the shaft. Gripper part 2 couples that shaft to a sponge, which is the gripping face: soft contact instead of a hard jaw. The IR sensors are mounted horizontally on gripper part 1 so the beam crosses the sponge gap and answers whether an object is there to grip. The time-of-flight sensor on the same part faces the ground and answers height. Treat the sponge as a consumable. Loom the sensor cables with the tool-motor cable so they cannot reach a pulley.

Console bay. White open box, joystick PCB screwed to the lid, shield inside, USB leaving the front, bay resting on the supply. Leave VMOT wires short and ferruled. Mains stays inside the supply enclosure. This document does not open or modify the supply.

13. Bring-up order

1. Unplug VMOT. Upload the jog sketch. Confirm joystick raw values sit near 512 at rest and swing toward 0 and 1023.
2. Set every A4988 Vref with the motor unplugged. Fit drivers. Common GND between supply and Uno.
3. Connect one motor. Apply 12 V. Jog that axis only. If it chatters, lower Vref or lengthen the step interval. If it runs backward, swap DIR.
4. Repeat for Y, then Z, then the tool motor. Do not enable all four at full current on the first power-up.
5. With the stick released, the table must not creep. Widen the dead zone if it does.
6. Confirm the time-of-flight reading falls as the carriage approaches the ground, and that a downward jog stops at the floor.
7. Confirm the IR bit changes when an object enters the gripper and that pickup() does nothing when the bit says empty.
8. Confirm the AprilTag 36h11 locks to a cell, that k matches the Trigonometry panel, and that a 7-cell-up gripper offset puts the jaws on the named working cell.
9. Add a series emergency switch on the 12 V feed before any unattended move. The joystick switch is a mode control, not a safety rated stop.

14. What to publish for certification

OSHA certification needs the design files, not only this narrative. Minimum set to put in a public repository, then submit:

- This document (CC BY-SA 4.0).
- STL or STEP of every printed bracket, the carriage, and the joystick bay, plus the original CAD if it exists (CERN-OHL-S-2.0).
- The jog sketch (.ino) and a one-page pin map.
- This bill of materials.
- The machine record in Section 6.
- A short note listing third-party modules and linking their datasheets (Arduino Uno, A4988, CNC Shield V3).

Recommended mark, once a UID is issued: “Open Source Hardware, CERN-OHL-S-2.0, certified open source hardware — [UID]”. Do not put the OSHWA mark on the machine before certification.

15. Safety and scope

Steppers and the 12 V feed can pinch and can overheat a mis-set driver. Mains wiring stays inside the listed supply. No claim is made of conformity to any product-safety standard. This is prototype documentation of one machine, version 0.1, for study and reproduction.

16. Certification checklist

Design files and this narrative allow study, modification, making, and sale. Non-commercial and no-derivatives terms are not used.

Original contribution is identified separately from third-party modules that already have public datasheets.

Bill of materials, machine record, pin map, bring-up order, and `hos_trigonometry_1_5.ino` are included.

Publish the STL or STEP of the printed nodes next to this document. Suggested repo path: Hardware/Trigonometry-1.5 in github.com/Aarav-111/Humanoid-Operating-System.

Suggested certification name: HOS Trigonometry 1.5. Submit only after the repo copy is public. Do not apply the OSHWA mark before a UID is issued.

Documentation licence: CC BY-SA 4.0. Hardware contribution licence: CERN-OHL-S-2.0. Upstream project: github.com/Aarav-111/Humanoid-Operating-System. HOS Trigonometry 1.5.